User Guide

A complete engineering reference for all 11 wizard steps — calculations, equations, standards, and best practice for drinking-water pump station design.

Document version **1.0** · May 2026 · For use with ALLL WPS Designer

Table of Contents

STEP 1

Project Setup

PURPOSE

Step 1 captures the administrative and configuration information that applies to the entire design. All subsequent calculations and exported reports use this data for traceability. Selecting the correct unit system here ensures consistent presentation throughout every step.

INPUT FIELDS

FIELD	REQUIRED	DESCRIPTION
Project Name	Yes	A unique, descriptive title used in all exports and the project dashboard (e.g., "Riverside PS Upgrade"). Maximum 120 characters.
Client	No	Name of the water authority, municipality, or owner for which the design is prepared.
Job Number	No	Internal reference number (e.g., WPS-2024-001).
Date	No	Design date; defaults to today. Appears on all exported reports.
Engineer	No	Engineer of record name, credential (e.g., "J. Smith, P.E.").
Notes	No	Free-text area for design intent, applicable standards, or scope limitations.

UNIT SYSTEM

SI (metric) — default

Flow: m^3/h · Head: m · Diameter: mm · Pressure: kPa

Recommended for international projects and new designs.

US Customary

Flow: gpm · Head: ft · Diameter: in · Pressure: psi

Use when the project specification is US-standard.

Enable **Show both units** to display SI and US values side-by-side in all result panels — useful for multi-disciplinary teams.

OPTIONAL MODULES

Include Surge Analysis — When enabled, a dedicated Water Hammer step (Step 9) is added to the wizard. Surge analysis uses the Joukowsky equation for rapid transient checks (Mode A) and a Method of Characteristics (MOC) numerical solver for detailed wave propagation (Mode B). Uncheck this if the design includes only gravity mains or if surge mitigation has been addressed by a specialist separately.

DESIGN FLOW RATE (Q DESIGN)

The single design-point flow rate `Q` used throughout all hydraulic calculations. Enter this value in the unit system selected above; the application converts internally and always stores SI values (m^3/h) for calculations.

> **Tip:** The design flow is the maximum continuous duty flow, not the instantaneous peak. For potable water stations, this is typically the maximum day demand (MDD) or the peak hour demand (PHD) as specified in the hydraulic brief.

VALIDATION

- Project Name must not be empty before advancing to Step 2.
- Design flow must be a positive number greater than zero.

STEP 2 System Nodes

PURPOSE

System nodes define the hydraulic boundary conditions at the suction source (upstream) and the delivery point (downstream). The elevation difference drives the static head component of Total Dynamic Head (TDH), while the boundary pressures represent residual pressure requirements that must be maintained.

GOVERNING EQUATION — BERNOULLI ENERGY EQUATION

BERNOULLI ENERGY EQUATION (SIMPLIFIED, PUMP STATION FORM)

$$TDH = (z_2 - z_1) + (p_2 - p_1)/(\rho g) + h\_f,total + h\_m,total$$

z_1 = upstream (suction) elevation [m]
z_2 = downstream (delivery) elevation [m]
p_1 = upstream boundary pressure [Pa]
p_2 = downstream residual pressure [Pa]
ρ = water density ≈ 998.2 kg/m³ at 20 °C
g = gravitational acceleration = 9.81 m/s²
h_f = pipe friction head loss [m]
h_m = minor head loss [m]

The static component is calculated immediately as you type:

STATIC HEAD

$$H\_static = z_2 - z_1 \ [m]$$

Positive value means pumping uphill (normal). Negative means pumping downhill (gravity-assisted).

INPUT FIELDS

NODE	FIELD	DESCRIPTION
Upstream (US)	Elevation	Hydraulic datum elevation at the wet well, reservoir, or suction source (m or ft above datum). Use the same datum throughout the project.
Upstream (US)	Pressure	Residual pressure at the upstream boundary. For a free-surface reservoir or wet well open to atmosphere, enter 0 kPa (0 psi).
Downstream (DS)	Elevation	Elevation at the delivery point — typically the top of the receiving reservoir, overhead tank, or zone entry point.
Downstream (DS)	Pressure	Minimum residual pressure required at the delivery point (e.g., 140 kPa / 20 psi for a distribution zone entry).

VALIDATION

- Both elevation fields must contain valid numbers before advancing.
- Negative static head is permitted (pumping downhill to pressurised zone).
- Boundary pressures default to 0 and are optional.

> **Note:** Use a consistent elevation datum (e.g., AHD, MSL, or local) across all nodes and pipeline profiles. Mixing datums is a common source of error that this tool cannot detect automatically.

`Ref: Bernoulli (1738)` `Ref: AWWA M11 §2` `Ref: Pump Handbook, Karassik et al., Ch. 8`

STEP 3

Suction Pipeline

PURPOSE

Define the pipe geometry, material, and fittings on the suction (inlet) side of the pump. The application calculates Darcy-Weisbach friction head loss and K-coefficient minor losses for each segment and aggregates them into the suction head loss used in TDH assembly.

DARCY-WEISBACH FRICTION LOSS

DARCY-WEISBACH EQUATION

$$h_f = f \cdot (L/D) \cdot V^2/(2g)$$

h_f = head loss due to friction [m]
f = Darcy-Weisbach friction factor (dimensionless) — computed by Colebrook-White iteration
L = pipe length [m]
D = internal pipe diameter [m]
V = mean flow velocity = $Q / (\pi D^2 / 4)$ [m/s]
g = 9.81 m/s^2

COLEBROOK-WHITE FRICTION FACTOR

COLEBROOK-WHITE EQUATION (IMPLICIT, SOLVED ITERATIVELY)

$$1/\sqrt{f} = -2.0 \log_{10}(\varepsilon/(3.7D) + 2.51/(Re\cdot\sqrt{f}))$$

ε = pipe roughness [m] — material-dependent (see table below)
D = diameter [m]
Re = Reynolds number = $\rho \cdot V \cdot D/\mu = V \cdot D/\nu$
ν = kinematic viscosity ≈ 1.004×10^{-6} m^2/s at 20 °C
The Swamee-Jain approximation is used as the initial estimate; Colebrook is iterated to convergence ($< 1 \times 10^{-6}$ tolerance).

Default Roughness Values (ε)

MATERIAL	E (MM)	NOTES
PVC / uPVC	0.0015	New; smooth interior
HDPE PE100	0.007	New; flexible, fused joints
Ductile Iron (DICL)	0.12	New, cement-lined
Cast Iron (unlined)	0.26	Old; may increase with age
Steel (welded)	0.046	New; commercial finish

Galvanised Iron	0.15	
Fiberglass / GRP	0.003	Smooth interior
Concrete / RCCP	0.30–3.0	Varies with finish; designer to confirm
Asbestos Cement	0.03	Old stock may increase
Copper	0.0015	Drawn, new

MINOR LOSSES (K-COEFFICIENT METHOD)

MINOR HEAD LOSS — K-COEFFICIENT

$$h_m = \Sigma K \cdot V^2/(2g)$$

K = fitting loss coefficient (dimensionless) — select from the built-in fittings library
V = velocity at the fitting location [m/s]
ΣK = sum of all fitting K values in the segment

Common K-values (representative; actual values depend on manufacturer and installation):

Gate valve (fully open): K ≈ 0.1
Globe valve (fully open): K ≈ 10
Ball valve (fully open): K ≈ 0.05
Butterfly valve: K ≈ 0.3–1.5
Check valve (swing): K ≈ 2.0

90° elbow (std): K ≈ 0.9
45° elbow (std): K ≈ 0.4
Tee (branch flow): K ≈ 1.8
Tee (run flow): K ≈ 0.4
Reducer (gradual): K ≈ 0.05–0.1

SEGMENT MANAGEMENT

- Add multiple segments to model pipes of different diameters or materials in series.
- Each segment requires: material, nominal diameter, and length.
- Use the **Accessories Picker** within each segment to add valves, elbows, and other fittings.
- Segments can be reordered using the up/down arrows.

VALIDATION

- At least one segment with diameter > 0 and length > 0 is required.
- Velocity in each segment is computed automatically and flagged if it exceeds AWWA M11 guidance (3.0 m/s for suction).

`Ref: Darcy (1857) / Weisbach (1845)` `Ref: Colebrook (1939)` `Ref: Moody (1944)` `Ref: AWWA M11 §4`

STEP 4 Clear Well Sizing

PURPOSE

The clear well (storage tank) provides the usable storage volume between the low water level (LWL) and the high water level (HWL). It must be large enough to satisfy minimum detention time requirements for disinfectant contact, limit pump starts per hour (cycling rate), and meet fire reserve and emergency storage obligations.

GOVERNING CALCULATIONS

Useful Storage Volume

CYLINDRICAL TANK

$$V_{useful} = \pi \cdot (D/2)^2 \cdot (HWL - LWL)$$

D = internal diameter [m]
HWL = high water level elevation [m]
LWL = low water level elevation [m]

RECTANGULAR TANK

$$V_{useful} = L \cdot W \cdot (HWL - LWL)$$

L = internal length [m], W = internal width [m]

Pump Cycling (Maximum Starts per Hour)

PUMP CYCLING — MAXIMUM STARTS PER HOUR

$$N_{starts} = Q_{pump} / (4 \cdot V_{useful}) \text{ [cycles/h]}$$

Q_pump = pump duty flow [m³/h]
V_useful = useful volume [m³]
This formula gives the theoretical maximum starts per hour at the worst-case flow ratio.
AWWA M32 and Ten States Standards recommend ≤ 6 starts/hour for most centrifugal pumps.

Detention Time

HYDRAULIC DETENTION TIME (HRT)

$$t_{det} = V_{total} / Q_{design} \text{ [h]}$$

V_total = total tank volume (including dead storage) [m³]
Q_design = design throughput flow rate [m³/h]

Minimum HRT for CT compliance: typically ≥ 0.5 h (30 min) per chlorine contact time requirements — verify with the applicable disinfection standard.

INPUT FIELDS

FIELD	DESCRIPTION
Tank geometry	Cylindrical or rectangular; affects volume formula used.
Internal dimensions	Diameter (cylindrical) or Length × Width (rectangular), plus total depth.
LWL / HWL	Low and High Water Levels as elevations or depths from base. HWL − LWL is the useful (operating) depth.
Pump duty flow	Flow pumped into or out of the tank. Used to calculate cycling rate.
Fill / draw flow	The net inflow during filling and outflow during drawdown cycles.

WHAT THE TOOL REPORTS

- **Useful volume** [m³] — storage between LWL and HWL.
- **Total volume** [m³] — full capacity (base to top).
- **Starts per hour** — cycling rate; flagged if > 6/h.
- **Detention time** — flagged if < 30 minutes.
- **Volume-elevation table** — tabulated volume vs. water level for operational reference.

`Ref: AWWA M11 — Steel Pipe Design` `Ref: AWWA M32 — Distribution System Requirements for Fire Protection`

`Ref: Ten States Standards (Recommended Standards for Water Works), §§ 5.3, 5.4`

STEP 5 Pump Selection & Curves

PURPOSE

Select the pump type, staging configuration, and variable-frequency drive (VFD) option. Enter the manufacturer's H-Q (head-flow), efficiency, and power curves so the tool can locate the operating point where the system curve intersects the pump curve.

PART A — PUMP TYPE & CONFIGURATION

FIELD	DESCRIPTION
Pump type	Centrifugal (end-suction, split-case, vertical turbine, submersible) or positive displacement. Most drinking-water stations use centrifugal.
Number of duty pumps	Pumps operating simultaneously in parallel at design flow.
Number of standby pumps	Redundant units on hot standby. AWWA and Ten States Standards typically require N+1 redundancy for critical supply.
VFD enabled	When checked, affinity-law scaling controls are shown. VFD allows speed reduction to match reduced demand flows.

PART B — H-Q CURVE ENTRY

Enter the pump manufacturer's head-flow data as a table of (Q, H) pairs — minimum three points are required for reliable interpolation. Include at least the shut-off point (Q = 0) and the runout (maximum flow) point.

Important: H-Q curves are for a single pump at rated speed. If multiple duty pumps are specified, the application automatically constructs the parallel-pump composite curve (flow adds at equal head).

AFFINITY LAWS (VFD OPERATION)

When a Variable Frequency Drive is in use, the pump operating point moves along the affinity law parabola as speed changes:

AFFINITY LAWS — CENTRIFUGAL PUMPS

$$Q_2/Q_1 = n_2/n_1$$

$$H_2/H_1 = (n_2/n_1)^2$$

$$P_2/P_1 = (n_2/n_1)^3$$

n_1 = rated speed [rpm], n_2 = reduced speed [rpm]

Q = flow rate [m^3/h], H = head [m], P = shaft power [kW]

These laws apply exactly for geometrically similar conditions; actual pump efficiency varies.

EFFICIENCY & POWER

HYDRAULIC POWER & EFFICIENCY

$$P_{hydraulic} = \rho \cdot g \cdot Q \cdot H \text{ [W]}$$

$$P_{shaft} = P_{hydraulic} / \eta_{pump} \text{ [W]}$$

$$P_{input} = P_{shaft} / \eta_{motor} \text{ [kW]}$$

$\rho = 998.2$ kg/m³, $g = 9.81$ m/s², Q in m³/s
η_{pump} = pump hydraulic efficiency (from manufacturer curve, 0–1)
η_{motor} = motor efficiency (typically 0.92–0.96)

Ref: HI 9.6.3 — Rotodynamic Pumps for Design and Application
Ref: Pump Handbook, Karassik et al., 4th ed.
Ref: AWWA M49 — Butterfly Valves (for selection guidance)

STEP 6 Discharge Pipeline

PURPOSE

Define the pipe network on the discharge (outlet) side of the pump — from the pump discharge flange to the delivery node. The same friction and minor-loss equations used in Step 3 apply here. Discharge pipeline losses typically dominate the system head for longer rising mains.

GOVERNING EQUATIONS

The Darcy-Weisbach and Colebrook-White equations documented in Step 3 apply identically to discharge segments. Refer to Step 3 for equation details, roughness values, and K-coefficient guidance.

RECAP — DISCHARGE HEAD LOSS

$$h_{f,discharge} = \Sigma \left[f_i \cdot (L_i/D_i) \cdot V_i^2/(2g) \right] + \Sigma K_i \cdot V_i^2/(2g)$$

Summation is over all discharge segments i.

VELOCITY GUIDANCE

PIPE LOCATION	RECOMMENDED RANGE	MAXIMUM (AWWA M11)
Suction main	0.6 – 1.5 m/s	3.0 m/s
Discharge main	0.9 – 2.5 m/s	3.5 m/s
Distribution mains	0.3 – 2.0 m/s	3.0 m/s

Design tip: High velocities increase friction losses and surge pressures; low velocities may cause sediment deposition and stagnation in potable water mains. Target 1.0–2.0 m/s in the discharge main for most drinking-water applications.

DISCHARGE-SIDE FITTINGS

The discharge side typically includes: non-return (check) valve, isolating gate/butterfly valve, pressure gauge connections, air release valves at high points (not modelled as a fitting loss, but important for system design), and the delivery connection. Use the Accessories Picker to account for each fitting's K-value.

VALIDATION

- At least one discharge segment with positive diameter and length is required before advancing to Step 7.
- Computed velocities are shown per segment; red highlighting appears if AWWA limits are exceeded.

Ref: Darcy-Weisbach (as Step 3)   Ref: AWWA M11 §4 — Velocity Limits

STEP 7 Hydraulic Results

PURPOSE

Step 7 assembles all the head components from Steps 2–6 into the Total Dynamic Head (TDH), computes the pump duty point, checks velocity compliance, and evaluates Net Positive Suction Head available (NPSHa) against the pump's required NPSHr.

TOTAL DYNAMIC HEAD (TDH) ASSEMBLY

TDH — FULL EXPRESSION

```
TDH = H_static + H_pressure + h_f,suction + h_m,suction +
h_f,discharge + h_m,discharge
```

H_static = $z_2 - z_1$ [m]

H_pressure = $(p_2 - p_1) / (\rho g)$ [m] — converted from kPa

h_f,suction = friction head loss, suction pipeline [m]

h_m,suction = minor head loss, suction fittings [m]

h_f,discharge = friction head loss, discharge pipeline [m]

h_m,discharge = minor head loss, discharge fittings [m]

The **Results Strip** at the top of the screen always shows the current TDH and flow rate regardless of which step is active.

NET POSITIVE SUCTION HEAD AVAILABLE (NPSHA)

Cavitation occurs when the suction-side absolute pressure drops below the liquid's vapour pressure. The available NPSH must exceed the pump manufacturer's required NPSH with an adequate safety margin.

NPSHA — AVAILABLE NET POSITIVE SUCTION HEAD

```
NPSHa = (p_atm - p_vapour)/(ρg) + z_s - h_f,s - h_m,s
```

p_atm = atmospheric pressure [Pa] (standard: 101 325 Pa at sea level)

p_vapour = vapour pressure of water at operating temperature [Pa]

= 2338 Pa at 20 °C; increases significantly with temperature

z_s = vertical distance from suction water level to pump centreline [m]

negative if pump is above water surface

h_f,s = suction friction head loss [m]

h_m,s = suction minor head loss [m]

HI 9.6.3 — MINIMUM NPSH MARGIN

```
NPSHa ≥ 1.1 × NPSHr (minimum 10 % margin)
```

NPSHr = pump manufacturer's required NPSH at duty flow [m]
A margin of 1.3–1.5× is recommended for pump longevity and reduced cavitation erosion.

VELOCITY CHECKS

All segment velocities computed in Steps 3 and 6 are summarised here with pass/fail status against AWWA M11 limits. Segments exceeding limits are highlighted in amber or red.

POWER ESTIMATE

PUMP DUTY POWER

$$P\_duty = \rho \cdot g \cdot Q \cdot TDH / \eta\_pump / \eta\_motor / 1000 \text{ [kW]}$$

Uses efficiency values from the pump curve data entered in Step 5 at the duty flow Q_design.

`Ref: HI 9.6.3 — NPSH Margin` `Ref: AWWA M11 §4` `Ref: Pump Handbook, Karassik et al.`

STEP 8 System Curve & Operating Point

PURPOSE

The system curve describes how system head varies with flow across the operating range. Overlaying it on the pump H-Q curve graphically reveals the **operating point** — the flow and head at which the pump will actually run.

SYSTEM CURVE GENERATION

SYSTEM CURVE — HEAD AT ARBITRARY FLOW Q_I

$$H_{sys}(Q_i) = H_{static} + H_{pressure} + (h_f + h_m)|_{Q=Q_i}$$

The static + pressure terms are constant; the friction and minor loss terms scale as Q^2.
The tool evaluates this at 8–10 evenly spaced flow points from Q = 0 to Q = 1.5 × Q_design to produce the curve.

SCALING PROPERTY

QUADRATIC SCALING OF HEAD LOSSES

$$h_f(Q_i) = h_f(Q_{design}) \times (Q_i / Q_{design})^2$$

Because $h_f \propto V^2 \propto Q^2$, friction and minor losses scale with the square of the flow ratio.
This gives the system curve its characteristic upward parabolic shape above the static head.

LOCATING THE OPERATING POINT

The operating point is found at the intersection of the pump H-Q curve and the system curve. The tool numerically solves for the flow Q* at which `H_pump(Q*) = H_sys(Q*)` using interpolation on the tabulated curve data.

Interpreting the Chart

- **Operating point to the right of design flow** — pump runs faster than intended; may cause overloading. Check motor power rating.
- **Operating point to the left of design flow** — pump may be oversized or system head underestimated.
- **Multiple intersections** — can occur with a flat or humped pump curve; the rightmost stable intersection is the operating point.
- **No intersection** — system head exceeds pump shut-off head; the pump cannot deliver flow. Increase pump size or reduce system head.

> **VFD Tip:** When VFD is enabled, affinity-law speed curves are plotted in light grey, showing how the operating point tracks along the system curve as speed reduces. This allows verification of minimum speed at minimum demand flow.

`Ref: HI 9.6.3 §3.6 — System Curve Analysis` `Ref: Pump Handbook, Karassik et al., Ch. 12`

STEP 9 Water Hammer & Surge Analysis

PURPOSE

Water hammer (surge) is the pressure wave generated when flow velocity changes rapidly — most commonly on sudden pump trip or valve closure. Uncontrolled surge can cause pipe bursts, joint failures, or column separation (negative pressure vacuum). Step 9 provides two levels of analysis:

- **Mode A — Joukowsky Quick Check**: Instant Joukowsky pressure rise, with Allievi slow-closure reduction for controlled valve operation. Suitable for initial feasibility and screening.
- **Mode B — Method of Characteristics (MOC)**: Full numerical transient simulation. Produces pressure envelopes, time histories, and wave-by-wave propagation. Suitable for detailed design and protection device sizing.

WAVE SPEED (HALLIWELL FORMULA)

PRESSURE WAVE SPEED — HALLIWELL THIN-WALL FORMULA

$$a = \sqrt{\frac{K/\rho}{1 + K \cdot D/(E_p \cdot e) \cdot C}}$$

a = wave speed [m/s]
K = bulk modulus of water ≈ 2.07 × 10^9 Pa
ρ = water density ≈ 998.2 kg/m³
D = internal pipe diameter [m]
E_p = pipe wall Young's modulus [Pa] — material dependent (see table)
e = pipe wall thickness [m]
C = pipe restraint factor:
1.0 for free (expansion joints throughout)
1 – ν/2 for anchored at upstream end
1 – ν² for fully restrained (buried)
ν = Poisson's ratio ≈ 0.3 for steel, 0.46 for HDPE

Typical Wave Speeds by Material

MATERIAL	E_P (MPA)	TYPICAL A (M/S)
HDPE PE100	900	300–400
PVC / uPVC	3 000	350–450
GRP / FRP	25 000	450–600
Ductile Iron (DICL)	168 000	900–1 100
Steel (welded)	206 000	1 000–1 300
Grey Cast Iron	96 500	800–1 000

Asbestos Cement	24 000	900–1 100
Concrete / RCCP	30 000	900–1 100

MODE A — JOUKOWSKY EQUATION

JOUKOWSKY INSTANTANEOUS PRESSURE RISE

$$\Delta H = a \cdot \Delta V / g$$

ΔH = surge head rise [m] — add to steady-state pressure
a = wave speed [m/s]
ΔV = change in flow velocity [m/s] (= V_steady for full pump trip)
$g = 9.81\ m/s^2$
This is the maximum surge for instantaneous stoppage (t_stop < 2L/a, the wave return period).

ALLIEVI SLOW-CLOSURE REDUCTION

ALLIEVI SLOW-CLOSURE FACTOR

$$\Delta H\_slow = \Delta H\_Joukowsky \times (2L/a) / t\_close$$

L = pipeline length [m]
a = wave speed [m/s]
2L/a = wave reflection period [s] — time for the pressure wave to travel to the reservoir and return
t_close = valve closure time [s]
When t_close > 2L/a, the surge is reduced; when t_close ≤ 2L/a it equals the Joukowsky value.

MODE B — METHOD OF CHARACTERISTICS (MOC)

MOC discretises the pipeline into space-time grid nodes and propagates pressure waves forward in time using the characteristic equations:

MOC CHARACTERISTIC EQUATIONS

$$C^+: H\_P = H\_A + (a/gA)\cdot(Q\_A - Q\_P) - R\cdot Q\_A\cdot|Q\_A|$$

$$C^-: H\_P = H\_B - (a/gA)\cdot(Q\_B - Q\_P) + R\cdot Q\_B\cdot|Q\_B|$$

H_P, Q_P = head and flow at the new time level at point P
H_A, Q_A = known conditions one timestep back at the upstream adjacent node
H_B, Q_B = known conditions one timestep back at the downstream adjacent node
A = pipe cross-sectional area [m^2]
R = friction resistance term = $f\cdot\Delta x/(2g\cdot D\cdot A^2)$
Δx = spatial reach length = a·Δt

MOC produces **pressure envelopes** (maximum and minimum head at every node), time histories at user-selected points, and checks all computed pressures against the pipe PN (pressure rating).

PROTECTION DEVICES

The tool allows specification of surge protection measures, which modify boundary conditions in the MOC solver:

- **Air vessel (surge tank)** — reduces upsurge by absorbing water volume
- **Slow-closing check valve** — limits reverse flow velocity on pump trip
- **Pressure relief valve** — vents excess pressure at the pump
- **Flywheel (pump inertia)** — slows pump rundown, reducing ΔV/Δt

`Ref: Joukowsky (1900)` `Ref: Allievi (1902)` `Ref: Wylie & Streeter, Fluid Transients in Systems (1993)` `Ref: Halliwell, A.R., Proc. ICE, 1963` `Ref: AWWA M11 §10 — Surge Pressure Design`

STEP 10 Engineering Checks

PURPOSE

Step 10 runs an automated code-compliance and best-practice review, comparing all computed values against industry-standard pass/fail criteria. No additional input is required — checks are derived from data entered in earlier steps and computed results.

REFERENCE STANDARDS APPLIED

- **AWWA M11** — Steel Pipe: A Guide for Design and Installation. Pipe velocity limits.
- **AWWA M32** — Distribution System Requirements for Fire Protection. Storage, redundancy, pressure requirements.
- **HI 9.6.3** — Rotodynamic Pumps for Design and Application. NPSH margin, duty point location, operating range.
- **Ten States Standards** (Recommended Standards for Water Works). Pump redundancy, clear well sizing, disinfectant contact time.

CHECK CATEGORIES

CATEGORY	CHECK	CRITERION
Velocity	Suction pipe velocity	≤ 3.0 m/s (AWWA M11)
Velocity	Discharge pipe velocity	≤ 3.5 m/s (AWWA M11)
Velocity	Minimum velocity (stagnation risk)	≥ 0.3 m/s at design flow
NPSH	NPSHa margin over NPSHr	≥ 1.1 × NPSHr (HI 9.6.3)
NPSH	Recommended margin	≥ 1.3 × NPSHr (best practice)
Clear Well	Pump starts per hour	≤ 6 starts/h (AWWA M32)
Clear Well	Minimum detention time	≥ 30 minutes
Pump Duty	Duty flow vs. pump BEP	Duty point within 70–120 % of BEP flow (HI 9.6.3)
Redundancy	Standby pump provision	N+1 minimum (Ten States §5.2)
Surge	Max surge pressure vs. pipe PN	Surge head + static ≤ 90 % of PN rating
Surge	Minimum pressure (vacuum / column sep.)	Minimum transient head ≥ −10 m (vapour pressure limit)

SEVERITY LEVELS

SEVERITY	COLOUR	MEANING
Critical	Red	Design does not meet minimum standard. Must be resolved before the design is finalised.
Warning	Amber	Design is marginal or does not meet best-practice recommendation. Review and document justification.
Pass	Green	Criterion satisfied.
Skipped	Grey	Check not applicable (e.g., surge checks when surge analysis is disabled).

Disclaimer: Engineering Checks are automated guides based on common industry standards and are not a substitute for professional engineering judgement. Results must be reviewed and verified by a qualified engineer. Local standards and project-specific requirements may impose stricter criteria.

STEP 11 Summary & Export

PURPOSE

Step 11 provides a complete summary of the design and allows the engineer to export the calculation report in various formats for documentation, peer review, client submission, or design-file archiving.

DESIGN SUMMARY PANEL

The summary panel collects the key outputs from all earlier steps into a single read-only view:

- Project metadata (name, client, job number, date, engineer)
- Design flow rate (m^3/h and gpm)
- TDH breakdown: static head, friction losses, minor losses, pressure differential
- Operating point: flow and head at pump curve intersection
- Pump power consumption (duty kW and annual kWh at rated hours)
- Clear well useful volume, cycling rate, and detention time
- Surge summary: maximum and minimum pressures, wave speed
- Engineering checks summary: total pass / warning / critical counts

EXPORT OPTIONS

FORMAT	CONTENTS	TYPICAL USE
PDF Report	Formatted calculation report with all inputs, equations, results, charts, and engineering check table.	Client submission, design filing, peer review.
Excel / CSV	Tabulated inputs and results in structured spreadsheet format for further analysis or QA checking.	Independent checking, asset management systems.
JSON (Save)	Machine-readable project file containing all input data. Can be loaded back into WPS Designer to resume or modify the design.	Project archiving, version control, reuse.

SAVING YOUR PROJECT

Use the **Save** button in the top toolbar to synchronise your project to the server. Saved projects appear in the Projects dashboard and can be loaded from any device. The application also retains the last project in browser local storage as a session backup.

LOADING SAMPLE PROJECTS

The **Load Sample** dropdown in the top toolbar provides three pre-built reference designs:

- **Basic Rising Main** — a standard single-stage pump station with a simple suction and discharge rising main.
- **Variable-Speed Transfer** — a VFD-controlled pump set with affinity-law speed variation.

- **Booster Station** — a mid-system booster with elevated suction pressure and short discharge to a high-pressure zone.

Loading a sample overwrites the current design — save your work first if needed.

> **Best practice for reports:** Before generating the final report, return to Step 10 (Engineering Checks) to confirm all critical checks pass. Amend the design as required, then export from Step 11. Include the engineering checks table in all client submissions.

App. Quick-Reference: Equations & Standards

CORE HYDRAULIC EQUATIONS

DARCY-WEISBACH (FRICTION HEAD LOSS)

$$h_f = f \cdot (L/D) \cdot V^2/(2g)$$

COLEBROOK-WHITE (FRICTION FACTOR)

$$1/\sqrt{f} = -2.0 \log_{10}(\varepsilon/(3.7D) + 2.51/(Re \cdot \sqrt{f}))$$

MINOR LOSSES (K-COEFFICIENT)

$$h_m = \Sigma K \cdot V^2/(2g)$$

TDH ASSEMBLY

$$TDH = H_{static} + H_{pressure} + h_{f,suction} + h_{m,suction} + h_{f,discharge} + h_{m,discharge}$$

NPSHA

$$NPSHa = (p_{atm} - p_{vapour})/(\rho g) + z_s - h_{f,s} - h_{m,s}$$

JOUKOWSKY WATER HAMMER

$$\Delta H = a \cdot \Delta V / g$$

WAVE SPEED (HALLIWELL)

$$a = \sqrt{ K/\rho / (1 + K \cdot D \cdot C/(E_p \cdot e)) }$$

AFFINITY LAWS (VFD)

$$Q_2/Q_1 = n_2/n_1 \quad \cdot \quad H_2/H_1 = (n_2/n_1)^2 \quad \cdot \quad P_2/P_1 = (n_2/n_1)^3$$

KEY STANDARDS REFERENCED

STANDARD	TOPIC
AWWA M11	Steel Pipe — Design and Installation; velocity limits, surge design
AWWA M32	Distribution System Requirements for Fire Protection; storage, redundancy
AWWA M49	Butterfly Valves; selection and application
HI 9.6.3	Rotodynamic Pumps — Design and Application; NPSH, operating point
Ten States Standards	Recommended Standards for Water Works; N+1 redundancy, clear well sizing
Moody (1944)	Friction Factor Chart; reference for Darcy-Weisbach regime mapping
Colebrook (1939)	Turbulent Flow in Pipes — implicit friction factor equation
Joukowsky (1900)	Über den hydraulischen Stoss — water hammer pressure formula
Allievi (1902)	Teoria del colpo d'ariete — slow-closure surge reduction
Halliwell (1963)	Proc. ICE — wave speed in thin-walled pipes
Wylie & Streeter (1993)	Fluid Transients in Systems — MOC numerical method
Karassik et al.	Pump Handbook, 4th ed. — centrifugal pump theory and selection

PHYSICAL CONSTANTS USED IN CALCULATIONS

CONSTANT	VALUE	UNITS
Gravitational acceleration (g)	9.81	m/s^2
Water density (ρ) at 20 °C	998.2	kg/m^3
Kinematic viscosity (ν) at 20 °C	1.004×10^{-6}	m^2/s
Dynamic viscosity (μ) at 20 °C	1.002×10^{-3}	Pa·s
Bulk modulus of water (K) at 20 °C	2.07×10^{9}	Pa
Vapour pressure (p_v) at 20 °C	2 338	Pa
Atmospheric pressure (p_atm) sea level	101 325	Pa

ALLL WPS Designer User Guide — Version 1.0, May 2026.